between 20% (too dry) and 80% (saturated), with optimal levels around 40-60%. Understanding this range helps you select sensors with appropriate measurement spans and avoid paying for unnecessary precision.

Finally, consider the rate of change—how quickly does this parameter fluctuate? Soil moisture changes gradually over hours or days, so sampling four times daily provides adequate coverage. In contrast, electrical power consumption can spike within seconds when appliances turn on, requiring more frequent sampling to capture these changes.

Standard questions to help define what you're measuring include:

- *What specific outcome or failure mode are you trying to detect or prevent?* Different failure modes require monitoring different parameters. For example, bearing failure in a motor might be detected through rising temperature, increased vibration, or abnormal acoustic patterns, each requiring different sensors and thresholds.
- *Which physical parameters change first when this condition occurs?* Early warning indicators give you time to plan maintenance, order parts, and schedule downtime whereas late-stage indicators only tell you when something has already changed. In an HVAC system, refrigerant pressure variations might signal compressor issues days before temperature control degrades, giving maintenance teams time to respond proactively rather than reactively.
- *How quickly does this parameter change during normal operation?* This determines your required sampling rate, data storage needs, and the complexity of your analysis algorithms. A wind turbine's blade vibration might need monitoring at thousands of samples per second to detect bearing defects, while monitoring a building's daily energy consumption pattern requires only hourly readings.
- *How much historical data do you need to establish baseline behavior?* Without a proper baseline, you can't distinguish normal variations from abnormal conditions. A manufacturing line might require three months of production data to understand typical variation patterns across different products, shifts, and seasonal conditions before anomaly detection becomes reliable.
- *What is the normal operating range for this parameter under typical conditions?* This defines the ranges of the thresholds you will monitor.